

SEAN SUN 孫祥恩

Image Algorithm Engineer | C++ / C# | Automated Optical Inspection (AOI)
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PROFESSIONAL SUMMARY

Image algorithm engineer with 3.5 years of experience developing and optimizing automated inspection algorithms for PCB defect detection, spanning image preprocessing, feature analysis, and multithreaded performance tuning. Brings an interdisciplinary foundation from a Master's in Biomedical Engineering, with prior research experience in 3D reconstruction, image processing, and motion capture for medical and rehabilitation applications. Proficient in C++/C# with working knowledge of Python, and experienced with engineering tools including AutoCAD, Creo, SolidWorks, and Abaqus, enabling independent, cross-disciplinary application development.

CORE SKILLS

- **Programming:** C++, C# (primary); Python (intermediate)
- **Image Processing & Algorithms:** image preprocessing, feature analysis, defect detection, 3D reconstruction, motion capture, biomechanics analysis
- **Software Engineering:** multithreading / performance optimization, Git version control
- **Engineering & Design Tools:** AutoCAD, Creo (Pro/ENGINEER), Fusion 360, SOLIDWORKS, SketchUp
- **Simulation:** Abaqus (Finite Element Analysis)
- **Hardware Platforms:** Kinect for Windows, Azure Kinect

PROFESSIONAL EXPERIENCE

Algorithm Engineer

UtechZone Co., Ltd.

- Designed and optimized automated inspection algorithms for detecting defects on printed circuit boards (PCBs), covering the full pipeline from image preprocessing to feature analysis.
- Applied image processing, data analysis, and linear algebra techniques to improve detection accuracy and robustness.
- Optimized multithreaded processing to improve inspection throughput and system performance.
- Used Git for version control in a collaborative development environment.
- Primary tools: C++, C#, Git

Full-time Research Assistant

National Taiwan University — Orthopaedic Engineering and Motion Analysis Laboratory

- Conducted research on 3D reconstruction, image processing, and motion capture, with applications in medical engineering and rehabilitation.
- Designed instructional materials and assessments for laboratory training.
- Developed an interactive motion evaluation platform application.
- Primary tools: JavaScript, Flutter

EDUCATION

National Taiwan University — Master's in Biomedical Engineering

2019 – 2021

National Taiwan University — Bachelor's in Mechanical Engineering

2015 – 2019